

Lecture 2: Euler-Maruyama Scheme, Gaussian Systems, and Models with Regime Switching

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Introduction to Data Assimilation
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An Example of DA Application



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US Air Force F-15 Jet Crashes IN Kuwait As Iran Steps Up Attacks On American Middle East Bases

- Pan, S., Su, H., Chu, J., & Wang, H. (2010). Applying a novel extended Kalman filter to missile–target interception with APN guidance law: A benchmark case study. *Control Engineering Practice*, 18(2), 159-167.
- Casasent, David, Charles P. Neuman, and John Lycas. "Optical Kalman filtering for missile guidance." *Applied optics* 23, no. 12 (1984): 1960-1966.

Outline

1. Euler-Maruyama Scheme
2. Gaussian and non-Gaussian systems
3. Complex models with regime switching

Euler-Maruyama Scheme

SDE Example: Setting and Notation

We consider the Itô SDE on $[0, T]$

$$dX_t = A(X_t, t) dt + B(X_t, t) dW_t, \quad X_0 \in L^2, \quad (1)$$

where W_t is a standard m -dimensional Wiener process and $X_t \in \mathbb{R}^d$.

We assume global Lipschitz and linear growth:

$$\|A(x, t) - A(y, t)\| + \|B(x, t) - B(y, t)\| \leq L\|x - y\|, \quad (2)$$

$$x^\top A(x, t) + \frac{1}{2}\|B(x, t)\|_F^2 \leq C(1 + \|x\|^2). \quad (3)$$

Time grid and Wiener increments

Let $0 = t_0 < t_1 < \dots < t_N = T$ be a uniform grid with

$$\Delta t = t_{n+1} - t_n$$

and

$$\Delta W_n := W_{t_{n+1}} - W_{t_n}.$$

Generating Wiener increments in practice: **Scalar increments** For scalar Brownian motion, the increments are independent Gaussians:

$$\Delta W_n \sim \mathcal{N}(0, \Delta t), \quad \Delta W_n = \sqrt{\Delta t} Z_n, \quad Z_n \sim \mathcal{N}(0, 1).$$

Multi-dimensional increments and Correlated Wiener

For m -dimensional W_t , take $Z_n \sim \mathcal{N}(0, I_m)$ and set

$$\Delta \mathbf{W}_n = \sqrt{\Delta t} Z_n$$

(independent components).

If a covariance $Q \in \mathbb{R}^{m \times m}$ is prescribed:

$$\Delta \mathbf{W}_n \sim \mathcal{N}(0, \Delta t Q), \quad \Delta \mathbf{W}_n = \sqrt{\Delta t} L Z_n,$$

with $Q = LL^\top$ (Cholesky).

Remark: Random seeds

Fix the RNG seed when comparing schemes to ensure identical noise realizations; this isolates the *discretization* error from sampling variability.

Euler–Maruyama (EM) and Convergence

The EM update is to solve (1) using:

$$X_{n+1} = X_n + A(X_n, t_n) \Delta t + B(X_n, t_n) \Delta W_n. \quad (4)$$

Definition (Strong and weak errors)

Let $X(t_n)$ be the exact solution. Strong error:

$$e_n^{\text{str}} := (\mathbb{E} \|X_n - X(t_n)\|^2)^{1/2}.$$

Strong order p :

$$\max_n e_n^{\text{str}} = O(\Delta t^p).$$

For smooth test φ ,

$$e_n^{\text{weak}} := |\mathbb{E} \varphi(X_n) - \mathbb{E} \varphi(X(t_n))|.$$

Weak order p iff

$$\max_n e_n^{\text{weak}} = O(\Delta t^p).$$

Strong and weak error convergence rates

Theorem (Moment bounds)

Under Lipschitz and linear growth, $\sup_{t \in [0, T]} \mathbb{E} \|X(t)\|^2 < \infty$ and for EM, $\max_n \mathbb{E} \|X_n\|^2 \leq C$ uniformly in Δt (for small enough Δt).

Theorem (Strong order of EM)

EM has strong order 1/2:

$$\max_{0 \leq n \leq N} (\mathbb{E} \|X_n - X(t_n)\|^2)^{1/2} \leq C \Delta t^{1/2}.$$

Theorem (Weak order of EM)

If A, B and test φ are sufficiently smooth with polynomial growth, then EM has weak order 1:

$$\max_n |\mathbb{E} \varphi(X_n) - \mathbb{E} \varphi(X(t_n))| \leq C \Delta t.$$

Gaussian and non-Gaussian systems

Gaussian system.

A system is called **Gaussian system** if it satisfies the following three conditions

- ▶ Gaussian initial data,
- ▶ linear dynamics, and
- ▶ Gaussian noise,

The simplest example is the scalar model

$$dx_t = (-ax_t + f_t)dt + \sigma dW(t).$$

1. Time-dependent moments of a Gaussian scalar model.

Consider the following 1-D linear-Gaussian equation with constant forcing f ,

$$dx_t = (-ax_t + f)dt + \sigma dW_t.$$

Using Reynold's decomposition

$$x_t = \langle x_t \rangle + x'_t, \quad \text{with } \langle x'_t \rangle = 0,$$

where $\langle \cdot \rangle$ represents the **ensemble average**, we can solve for the first and second normalized **central moments** (mean and variance)

$$\langle x_t \rangle \quad \text{and} \quad \langle (x'_t)^2 \rangle.$$

Besides, direct application of Ito's formula helps solve the first and second **moments**

$$\langle x_t \rangle \quad \text{and} \quad \langle x_t^2 \rangle.$$

The relation between $\langle x_t^2 \rangle$ and $\langle (x'_t)^2 \rangle$ is seen

$$\begin{aligned} (x'_t)^2 &= (x_t - \langle x_t \rangle)^2 = x_t^2 + \langle x_t \rangle^2 - 2x_t \langle x_t \rangle \\ \langle (x'_t)^2 \rangle &= \langle x_t^2 \rangle + \langle x_t \rangle^2 - 2\langle x_t \rangle^2 = \langle x_t^2 \rangle - \langle x_t \rangle^2. \end{aligned}$$

Using Reynold's decomposition to solve the central moments $\langle x_t \rangle$ and $\langle (x'_t)^2 \rangle$.

$$dx_t = (-ax_t + f)dt + \sigma dW_t. \quad (1)$$

$$x_t = \langle x_t \rangle + x'_t, \quad \text{with } \langle x'_t \rangle = 0.$$

Taking the ensemble average of (1) yields

$$d\langle x_t \rangle = (-a\langle x_t \rangle + f)dt \quad (2)$$

such that the evolution of the **mean** of x_t is given by

$$\langle x_t \rangle = \langle x_0 \rangle e^{-at} + \frac{f}{a}(1 - e^{-at}).$$

Subtracting (2) from (1) yields

$$dx'_t = -ax'_t dt + \sigma dW_t.$$

Use Ito's formula,

$$\begin{aligned} dx_t'^2 &= 2x'_t dx'_t + (dx'_t)^2 \\ &= \left(-2ax_t'^2 + \sigma^2 \right) dt + 2\sigma x'_t dW_t \end{aligned} \quad (3)$$

Therefore, taking the ensemble average of (3) yields

$$d\langle x_t'^2 \rangle = \left(-2a\langle x_t'^2 \rangle + \sigma^2 \right) dt,$$

which gives the **variance** of x_t

$$\langle x_t'^2 \rangle = \langle x_0'^2 \rangle e^{-2at} + \frac{\sigma^2}{2a}(1 - e^{-2at}).$$

2. Statistical equilibrium state and decorrelation time.

Gaussian model: $dx_t = (-ax_t + f)dt + \sigma dW_t,$

Mean: $\bar{x}_t = \langle x_0 \rangle e^{-at} + \frac{f}{a}(1 - e^{-at}),$

Variance: $Var(x_t) = \langle x_0'^2 \rangle e^{-2at} + \frac{\sigma^2}{2a}(1 - e^{-2at}).$

Statistical equilibrium state:

$$\bar{x}_\infty = \frac{f}{a}, \quad Var(x)_\infty = \frac{\sigma^2}{2a}.$$

What is a ? — the **damping** of the system.

The inverse of a , i.e., $1/a$ is the **decorrelation** time, which measures the memory of the system.

Definition: the decorrelation time of a stationary process is given by

$$\tau_{corr} = \int_0^\infty R(s)ds,$$

where $R(s)$ is the autocorrelation function,

$$R(s) = \frac{E[(x_t - \bar{x}_\infty)(x_{t+s} - \bar{x}_\infty)]}{Var(x)_\infty}.$$

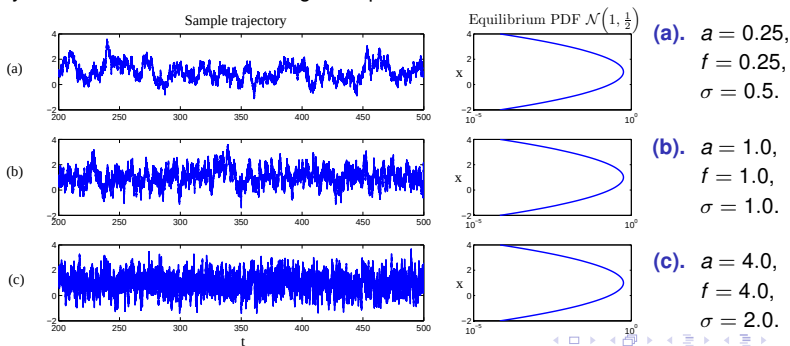
Recall the Gaussian system

$$dx_t = (-ax_t + f)dt + \sigma dW_\gamma.$$

The equilibrium mean, variance and decorrelation time are given by

$$\bar{x}_\infty = \frac{f}{a}, \quad \text{Var}(x)_\infty = \frac{\sigma^2}{2a}, \quad \tau_{\text{corr}} = \frac{1}{a}.$$

The system has three free parameters (a , f and σ) while the PDF has only two freedom (mean and variance). This implies the PDF is **not sufficient** to fully describe the dynamics. Consider the following example:



3. Fokker-Planck equation of the Gaussian scalar model.

Consider the scalar system

$$dx = (-ax + f)dt + \sigma dW_t, \quad \text{where } a, f \text{ and } \sigma \text{ are constant.}$$

The associated Fokker-Planck equation is

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x} \left((-ax + f)p \right) + \frac{\sigma^2}{2} \frac{\partial^2 p}{\partial x^2}. \quad (4)$$

The stationary solutions of (4) satisfy $\partial p / \partial t = 0$, which implies that

$$\left(x - \frac{f}{a} \right) p + \frac{\sigma^2}{2a} \frac{\partial p}{\partial x} = c_1.$$

The boundary conditions $p \xrightarrow{x \rightarrow \pm\infty} 0$ and $\partial p / \partial x \xrightarrow{x \rightarrow \pm\infty} 0$ imply that $c_1 = 0$. Thus, the stationary solution p_{eq} of (4) is Gaussian given by

$$p(x) = N_0 \exp \left(-\frac{1}{2} \frac{(x - f/a)^2}{\sigma^2/2a} \right), \quad (5)$$

with mean $\bar{x}_\infty = f/a$ and variance $\text{Var}(x)_\infty = \sigma^2/2a$.

4. Summary

- ▶ A system is called **Gaussian system** if it satisfies the following conditions: 1) Gaussian initial data, 2) linear dynamics, and 3) Gaussian noise.
- ▶ Use Reynold's decomposition and the Ito formula to solve the mean and variance of a Gaussian scalar system,

$$dx_t = (-ax_t + f)dt + \sigma dW_\gamma.$$

The equilibrium **mean**, **variance** and **decorrelation time** are given by

$$\bar{x}_\infty = \frac{f}{a}, \quad \text{Var}(x)_\infty = \frac{\sigma^2}{2a}, \quad \tau_{corr} = \frac{1}{a}.$$

- ▶ Use Fokker-Planck equation to solve the equilibrium distribution

$$p(x) = N_0 \exp \left(-\frac{1}{2} \frac{(x - f/a)^2}{\sigma^2/2a} \right). \quad (6)$$

Non-Gaussian system.

In this section, we will study the following two kinds of non-Gaussian system.

1. Linear scalar model with multiplicative noise

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + c dW_c(t),$$

where W_b and W_c are independent scalar Wiener processes.

2. Canonical scalar model with cubic nonlinearity and deterministic structural instability

$$dx = [F + ax + bx^2 - cx^3]dt + (A - Bx)dW + \sigma dW_A.$$

Normal form for a scalar reduced stochastic climate model for the low-frequency variability ².

²**A. J. Majda, C. Franzke and D. Crommelin**, Normal forms for reduced stochastic climate models, *PNAS*, **106**(10), 2009.

1. Linear scalar model with multiplicative noise.

$$du(t) = [-au(t) + f]dt + b u(t)dW_b(t) + c dW_c(t), \quad (7)$$

1.1. Exact path-wise solutions for the system

1. Solve the general solution of the homogeneous equation associated with (7),

$$du(t) = -a u(t)dt + b u(t)dW_b(t). \quad (8)$$

2. Find the special solution of the non-homogeneous equation.

In the following, we will compare the procedure of solving SDE (7) with that of solving the following ODE

$$du(t) = [-au(t) + f]dt. \quad (9)$$

$$\text{Ito's formula } d(f(X_t)) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2.$$

Step 1. Solution of homogeneous equation,

ODE:

$$du(t) = -a u(t)dt$$

$$d \ln u(t) = \frac{du(t)}{u(t)} = -adt$$

$$\begin{aligned} u(t) &= u_0 \exp \left(- \int_{t_0}^t a(s)ds \right) \\ &= u_0 \Psi(t). \end{aligned}$$

Linear non-Gaussian SDE:

$$du(t) = -a u(t)dt + b u(t)dW_b(t)$$

$$d \ln u(t) = \frac{du(t)}{u(t)} = -adt + b dW_b(t), \text{ Be careful!}$$

$$\begin{aligned} u(t) &= u_0 \exp \left(- \int_{t_0}^t a(s)ds + \int_0^t b(s)dW_b(s) \right) \\ &= u_0 \Phi(t) \end{aligned}$$

$$\text{Ito's formula } d(f(X_t)) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)(dX_t)^2.$$

Step 1. Solution of homogeneous equation,

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Linear non-Gaussian SDE:

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$$d \ln u(t) = \frac{du(t)}{u(t)} = -adt + b dW_b(t), \text{ Be careful!}$$

$$u(t) = u_0 \exp \left(- \int_{t_0}^t a(s)ds + \int_0^t b(s)dW_b(s) \right) \\ = u_0 \Phi(t)$$

This is wrong!

$$d \ln u \neq \frac{du(t)}{u(t)} \quad \text{when } u(t) \text{ is stochastic.}$$

Correct way: Set $f(u) = \ln u$ and use Ito's formula

$$d \ln u(t) = \frac{du}{u} - \frac{1}{2} \frac{(du)^2}{u^2} = - \left(a + \frac{1}{2} b^2 \right) dt + b dW_b(t).$$

$$u(t) = u_0 \exp \left(- \int_{t_0}^t (a(s) + \frac{1}{2} b(s)^2) ds + \int_{t_0}^t b(s) dW_b(s) \right) = u_0 \Phi(t).$$

Step 2. Solution of the non-homogeneous equation.

Solving ODE:

$$du(t) = -audt + fdt \quad \text{with } \Psi(t) = \exp\left(-\int_{t_0}^t a(s)ds\right)$$

$$d(u\Psi^{-1}) = du\Psi^{-1} + au\Psi^{-1}dt = (du + audt)\Psi^{-1} = f\Psi^{-1}dt$$

$$u(t) = \Psi(t) \int_{t_0}^t \frac{f(s)}{\Psi(s)} ds$$

Solving linear non-Gaussian SDE:

$$du(t) = -au(t)dt + bu(t)dW_b(t) + fdt + cdW_c(t)$$

$$\text{with } \Phi(t) = \exp\left(-\int_{t_0}^t (a(s) + \frac{1}{2}b(s)^2)ds + \int_0^t b(s)dW_b(s)\right).$$

Applying Ito's formula yields,

$$d(u\Phi^{-1}) = du\Phi^{-1} + u d(\Phi^{-1}) + dud(\Phi^{-1}),$$

where

$$d(\Phi^{-1}) = -\Phi^{-2}d\Phi + \frac{1}{2}\Phi^{-3}(d\Phi)^2 = ((a + b^2)dt - bdW_b(t))\Phi^{-1}.$$

Thus

$$du\Phi^{-1} = (-audt + budW_b(t))\Phi^{-1} + (fdt + cdW_c(t))\Phi^{-1}$$

$$ud\left(\Phi^{-1}\right) = (audt + b^2udt - budW_b(t))\Phi^{-1}$$

$$dud\left(\Phi^{-1}\right) = -b^2udt\Phi^{-1}$$

Combining the above equations yields

$$d\left(u\Phi^{-1}\right) = (fdt + cdW_c(t))\Phi^{-1}$$

Therefore, the solution of the non-homogeneous equation is given by

$$u(t) = \Phi(t) \left(\int_{t_0}^t \frac{f(s)}{\Phi(s)} ds + \int_{t_0}^t \frac{c(s)dW_c(s)}{\Phi(s)} \right).$$

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + cdW_c(t),$$

$$u(t) = u_h(t) + u_s(t),$$

$$u_h(t) = u_0 \exp \left(- \int_{t_0}^t (a(s) + \frac{1}{2}b(s)^2)ds + \int_{t_0}^t b(s)dW_b(s) \right) = u_0\Phi(t),$$

$$u_s(t) = \Phi(t) \left(\int_{t_0}^t \frac{f(s)}{\Phi(s)} ds + \int_{t_0}^t \frac{c(s)dW_c(s)}{\Phi(s)} \right).$$

1.2. Invariant measure for the linear non-Gaussian system (10)

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + c dW_c(t). \quad (10)$$

The invariant measure associated with (10) is given by the time-independent solutions of the Fokker-Planck equation

$$\partial_t p(u, t) = -\partial_u [(-au + f)p(u, t)] + \frac{1}{2}\partial_u^2 [(b^2 u^2 + c^2)p(u, t)] \equiv \mathcal{L}_{FP} p(u, t). \quad (11)$$

The stationary solutions of (11), i.e. $\mathcal{L}_{FP} p_{eq} = 0$, satisfy

$$\partial_u [(b^2 u^2 + c^2)p_{eq}(u)] = 2(-au + f)p_{eq}(u), \quad (12)$$

which can be easily integrated to yield

$$p_{eq}(u) = \frac{N_0}{A(u)} \exp \left(\int_{u_0}^u B(u')/A(u') du' \right)$$

where

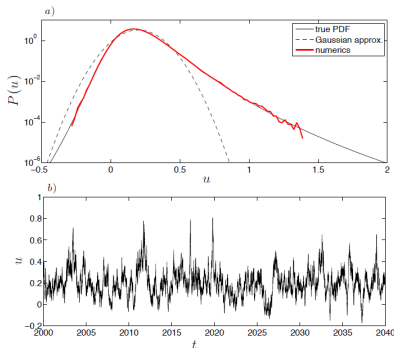
$$A(u) = bu^2 + c^2, \quad B(u) = 2(-au + f).$$

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + c dW_c(t).$$

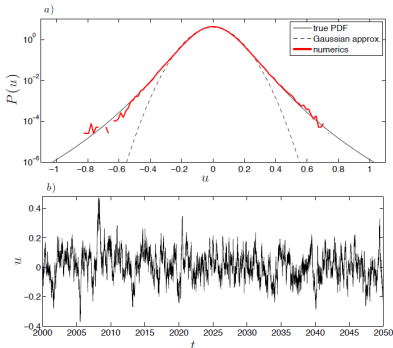
(1) $a > 0, b \neq 0, c \neq 0$ Non-zero additive and multiplicative noise; non-Gaussian PDF with **fat algebraic tails**.

$$p_{eq}(u) = \frac{N_0}{(b^2 u^2 + c^2)^{1+a/b^2}} \exp\left(\frac{2f}{|bc|} \arctan\left(\frac{|b|}{|c|} u\right)\right),$$

with N_0 a normalizing constant.



$(a = 5, b = 1, c = 0.3, f = 0.6.)$

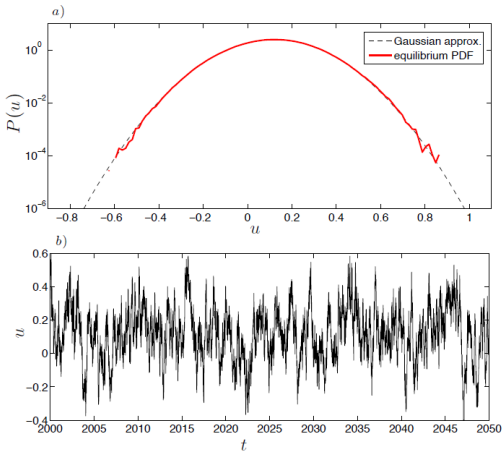


$(a = 5, b = 1, c = 0.5, f = 0.)$

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + c dW_c(t).$$

(2) $a > 0, b = 0, c \neq 0$ Only additive noise; Gaussian PDF.

$$p_{eq}(u) = N_0 \exp\left(-\frac{a}{c^2}(u - f/a)^2\right).$$



($a = 5, b = 0, c = 0.5, f = 0.6$.)

$$du(t) = [-au(t) + f]dt + bu(t)dW_b(t) + c dW_c(t).$$

(3) $a > 0, b = 0, c = 0$ Deterministic case; no noise.

$$p_{eq}(u) = \delta(u - f/a).$$

(4) $a > 0, b \neq 0, c = 0$ The invariant measure does not exist at $u = 0$.

$$p_{eq}(u) = \frac{N_0}{u^{2(1+a/b^2)}} \exp(-2f/(b^2 u)).$$

1.3. Time-dependent moments.

$$du(t) = [-a(t)u(t) + f(t)]dt + b u(t)dW_b(t) + c dW_c(t). \quad (7)$$

We assume here that $b = \text{const}$, $c = \text{const}$ and the forcing is bounded. We show that for time-periodic damping $a(t) = a_0 + \tilde{a}(t)$ with period T and a finite positive mean

$$0 < a_0 \equiv \frac{1}{T} \int_{t_0}^{t_0+T} a(s)ds,$$

the moments of the time-dependent PDFs of (7) exist up to order n provided that $c \neq 0$ and

$$-na_0 + \frac{1}{2}n(n-1)b^2(t-t_0) < 0. \quad (13)$$

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Remark: Time dependent moments can be used to approximate the time-dependent PDF.

How? Maximum entropy principle!

$$du(t) = [-a(t)u(t) + f(t)]dt + b u(t)dW_b(t) + c dW_c(t).$$

Ito's formula applied to u^n ,

$$du^n = nu^{n-1}du + \frac{n(n-1)}{2}u^{n-2}(du)^2 \quad (14)$$

$$\begin{aligned} du^n &= nu^{n-1} [(-au(t) + f)dt + b u(t)dW_b(t) + c dW_c] + \frac{n(n-1)}{2}u^{n-2}(b^2u^2 + c^2)dt \\ &= \left(-na(t) + \frac{n(n-1)}{2}b^2\right)u^n dt + \left(nf(t)u^{n-1} + \frac{n(n-1)}{2}c^2u^{n-2}\right)dt + bnu^n dW_b + nu^{n-1}cdW_c \end{aligned}$$

Taking the ensemble average leads to

$$d\langle u^n(t) \rangle = \left(\left(-na(t) + \frac{n(n-1)}{2}b^2\right) \langle u^n(t) \rangle + nf(t)\langle u^{n-1}(t) \rangle + \frac{n(n-1)}{2}c^2\langle u^{n-2}(t) \rangle \right) dt$$

Integrating the above equation yields

$$\langle u^n(t) \rangle = \langle u_0^n \rangle e^{J_n(t_0, t)} + \int_{t_0}^t e^{J_n(s, t)} \left(nf(s)\langle u^{n-1}(s) \rangle + \frac{1}{2}n(n-1)c^2\langle u^{n-2}(s) \rangle \right) ds \quad (15)$$

where

$$J_n(s, t) = -n \int_s^t a(t')dt' + \frac{1}{2}n(n-1)b^2(t-s). \quad (16)$$

In the above formulas we additionally set $\langle u(t)^{n-k} \rangle = 1$ for $n = k, k = 1, 2$

and $\langle u(t)^{n-k} \rangle = 0$ for $n - k < 0, k = 1, 2$.

2. Canonical scalar model with cubic nonlinearity and deterministic structural instability

$$dx = [F + ax + bx^2 - cx^3]dt + (A - Bx)dW + \sigma dW_A. \quad (17)$$

2.1. Equilibrium PDF

The Fokker-Planck equation associated with system (17) is given by

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}[(F + ax + bx^2 - cx^3)p] + \frac{1}{2} \frac{\partial^2}{\partial x^2}[(A - Bx)^2 + \sigma^2]p.$$

- For $A \neq 0$ and $B \neq 0$,

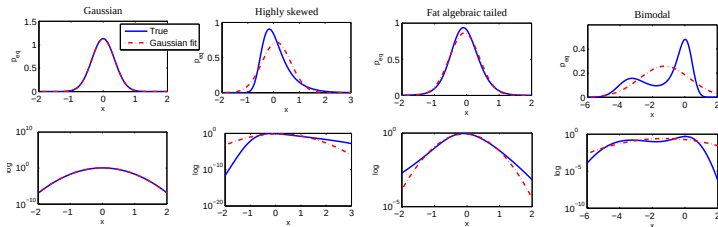
$$p_{eq}(x) = \underbrace{\frac{N_0}{((Bx - A)^2 + \sigma^2)^{a_1}} \exp\left(d \arctan\left(\frac{Bx - A}{\sigma}\right)\right)}_{\text{Fat algebraic tails}} \underbrace{\exp\left(-\frac{c_1 x^2 + b_1 x}{B^4}\right)}_{\text{Gaussian part}}.$$

- For $A = B = 0$,

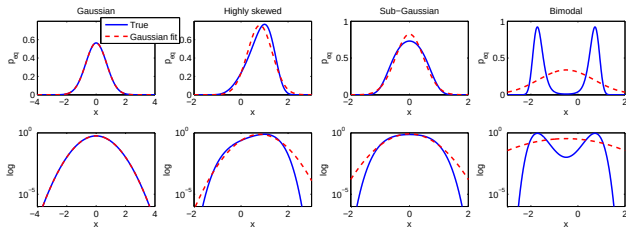
$$p_{eq}(x) = N_0 \exp\left(\frac{2}{\sigma^2}\left(Fx + \frac{a}{2}x^2 + \frac{b}{3}x^3 - \frac{c}{4}x^4\right)\right).$$

$$dx = [F + ax + bx^2 - cx^3]dt + (A - Bx)dW + \sigma dW_A.$$

$$p_{eq}(x) = \frac{N_0}{((Bx - A)^2 + \sigma^2)^{a_1}} \exp\left(d \arctan\left(\frac{Bx - A}{\sigma}\right)\right) \exp\left(-\frac{c_1 x^2 + b_1 x}{B^4}\right).$$



$$p_{eq}(x) = N_0 \exp\left(\frac{2}{\sigma^2}\left(Fx + \frac{a}{2}x^2 + \frac{b}{3}x^3 - \frac{c}{4}x^4\right)\right), \quad \text{with } A = B = 0.$$



2.2. Time-dependent moments?

$$dx = (F + ax + bx^2 - cx^3)dt + (A - Bx)dW + \sigma dW_A.$$

$$d\langle x \rangle = (F + a\langle x \rangle + b\langle x^2 \rangle - c\langle x^3 \rangle)dt$$

$$d\langle x^2 \rangle = \langle 2xdx \rangle + \langle \frac{1}{2}(dx)^2 \rangle$$

$$= \left((A^2 + \sigma^2) + (F - 2AB)\langle x \rangle + (a + B^2)\langle x^2 \rangle + b\langle x^3 \rangle - c\langle x^4 \rangle \right) dt$$

$$d\langle x^3 \rangle = \dots$$

Moment $\langle x^n \rangle$ depends on higher order moments $\langle x^{n+1} \rangle$ and $\langle x^{n+2} \rangle$.

The system is never closed!

How to approximate the leading a few moments? Closure methods.

Quasi-Gaussian closure for the cubic system:

- ▶ Only include the evolution of the mean and variance.
- ▶ The third order central moment is assumed to be zero.
- ▶ The fourth order central moments is expressed via the second central moments.

3. Summary

1. Linear scalar model with multiplicative noise

$$du(t) = [-au(t) + f]dt + b u(t)dW_b(t) + c dW_c(t).$$

- ▶ Path-wise solution (need Ito formula and be careful for the differential)
- ▶ Invariant measure (non-Gaussian features).
- ▶ Time dependent moments (useful for approximating time-dependent PDF).

2. Canonical scalar model with cubic nonlinearity and deterministic structural instability

$$dx = [F + ax + bx^2 - cx^3]dt + (A - Bx)dW + \sigma dW_A.$$

- ▶ Equilibrium PDF (non-Gaussian features).
- ▶ No closed form for time-dependent moments due to the nonlinearity.
- ▶ How to approximate time-dependent moments/PDF?

Regime switching

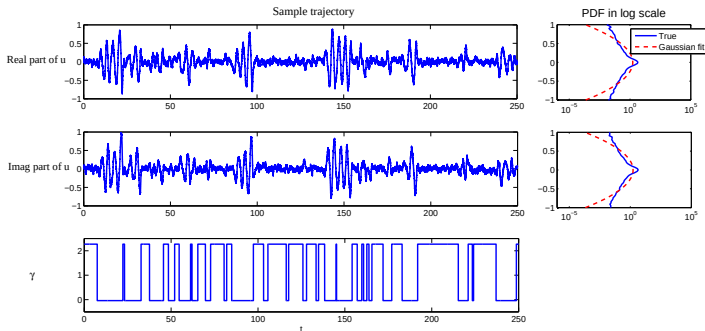
Building complex models with regime switching

$$du = [(-\gamma + i\omega_u)u + f(t)]dt + \sigma_u dW_u,$$

(18)

γ satisfies two-state Markov jump process.

- ▶ Two layer model, in which the unresolved variable is driven by a two-state Markov jump process.
- ▶ Two-state Markov jump process can introduce the intermittency.
- ▶ Example in AOS: Rossby wave with baroclinic instability³.



³Gershgorin et al, *JCP*, 32-57, 2010

1. Finite Markov chains

- ▶ Define finite state set $S = \{s_i\}_{i \in \mathcal{N}}$, $\mathcal{N} = \{1, 2, \dots, N\}$.

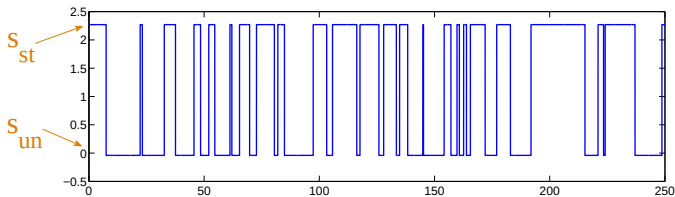
Non-trivial: at least one of the s_i is different from the rest.

- ▶ A **finite Markov chain** is a random process that undergoes transitions from one state to another on a state space.
- ▶ It must possess the **Markov property** (“memorylessness”): the probability distribution of the next state depends only on the current state and not on the sequence of events that preceded it.
- ▶ Doesn’t satisfy the Fokker-Planck equation but satisfies so-called **Master equation**.
- ▶ Has discontinuous paths.

2. Two-state Markov jump process:

Two assumptions:

1. Markov property: $P(X_t = y | X_r, 0 \leq r \leq \tau) = P(X_t = y | X_\tau)$,
2. Time homogeneity: $P(X_t = y | X_\tau = x) = P(X_{t-\tau} = y | X_0 = x)$.



Switching rate:

$$\nu : s_{st} \rightarrow s_{un},$$

$$\mu : s_{un} \rightarrow s_{st}.$$

Switching times:

$$P(T_{st} \leq t) = 1 - e^{-\nu t},$$

$$P(T_{un} \leq t) = 1 - e^{-\mu t}.$$

Expectation:

$$E[X] = \frac{\mu s_{st} + \nu s_{un}}{\nu + \mu}.$$

These rates define the following local transition probabilities for small Δt

$$P(X_{t+\Delta t} = s_{un} | X_t = s_{st}) = \nu \Delta t + o(\Delta t),$$

$$P(X_{t+\Delta t} = s_{st} | X_t = s_{un}) = \mu \Delta t + o(\Delta t),$$

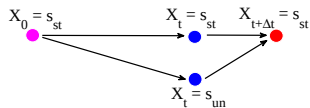
$$P(X_{t+\Delta t} = s_{st} | X_t = s_{st}) = 1 - \nu \Delta t + o(\Delta t),$$

$$P(X_{t+\Delta t} = s_{un} | X_t = s_{un}) = 1 - \mu \Delta t + o(\Delta t),$$

3. Master equation of two-state Markov jump process

We write the differential equation for the transition probability from stable to unstable regimes

$$p_t(s_{st}, s_{un}) = P(X_t = s_{un} | X_0 = s_{st}):$$



$$\begin{aligned} p_{t+\Delta t}(s_{st}, s_{un}) &= p_t(s_{st}, s_{un})p_{\Delta t}(s_{un}, s_{un}) + p_t(s_{st}, s_{st})p_{\Delta t}(s_{st}, s_{un}) \\ &= p_t(s_{st}, s_{un})(1 - \mu\Delta t) + p_t(s_{st}, s_{st})\nu\Delta t + o(\Delta t) \end{aligned}$$

After regrouping terms, we obtain a finite-difference equation

$$\frac{p_{t+\Delta t}(s_{st}, s_{un}) - p_t(s_{st}, s_{un})}{\Delta t} = -\mu p_t(s_{st}, s_{un}) + \nu p_t(s_{st}, s_{st}) + o(1).$$

In the limit $\Delta t \rightarrow 0$, we obtain the following differential equation with initial condition

$$\begin{aligned} \partial_t p_t(s_{st}, s_{un}) &= -\mu p_t(s_{st}, s_{un}) + \nu p_t(s_{st}, s_{st}), \\ p_t(s_{st}, s_{un})|_{t=t_0} &= 0. \end{aligned}$$

Let

$$\hat{P}_t = \begin{pmatrix} p_t(s_{st}, s_{st}) & p_t(s_{st}, s_{un}) \\ p_t(s_{un}, s_{st}) & p_t(s_{un}, s_{un}) \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} -\nu & \nu \\ \mu & -\mu \end{pmatrix}.$$

Then $\hat{P}_t$ satisfies

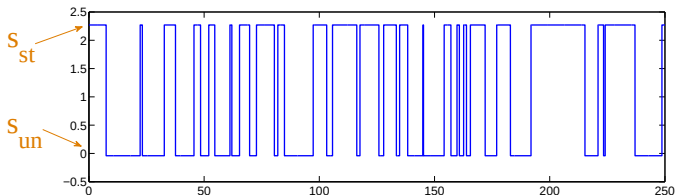
$$\frac{\partial}{\partial t} \hat{P}_t = \hat{P}_t A \quad \text{with} \quad \hat{P}_t|_{t=0} = I,$$

The master equation reads

$$\frac{\partial}{\partial t} \hat{P}_t = \hat{P}_t A \quad \text{with} \quad \hat{P}_t|_{t=0} = I,$$

the solution of which is given by

$$\begin{aligned} P_t(s_{st}, s_{st}) &= \frac{\mu}{\nu + \mu} + \frac{\nu}{\nu + \mu} e^{-(\nu + \mu)t}, & P_t(s_{un}, s_{st}) &= \frac{\mu}{\nu + \mu} \left(1 - e^{-(\nu + \mu)t}\right), \\ P_t(s_{un}, s_{un}) &= \frac{\nu}{\nu + \mu} + \frac{\mu}{\nu + \mu} e^{-(\nu + \mu)t}, & P_t(s_{st}, s_{un}) &= \frac{\nu}{\nu + \mu} \left(1 - e^{-(\nu + \mu)t}\right), \\ P_{eq}(s_{st}, s_{st}) &= P_{eq}(s_{un}, s_{st}) = \frac{\mu}{\nu + \mu}, \\ P_{eq}(s_{st}, s_{un}) &= P_{eq}(s_{un}, s_{un}) = \frac{\nu}{\nu + \mu}. \end{aligned}$$



$$\nu : s_{st} \rightarrow s_{un}, \quad \mu : s_{un} \rightarrow s_{st}.$$

4. Conditionally linear SDE with regime switching

$$du = [-\gamma(t)u + i\omega u + f(t)]dt + \sigma dW(t).$$

- $\gamma(t)$ real scalar diffusion process

$$\begin{aligned} u(t) = & e^{-\int_{t_0}^t \gamma(t')dt' + i\omega(t-t_0)} u_0 + \int_{t_0}^t f(s) e^{-\int_s^t \gamma(t')dt' + i\omega(t-s)} ds \\ & + \sigma \int_{t_0}^t e^{-\int_s^t \gamma(t')dt' + i\omega(t-s)} dW(s). \end{aligned}$$

Non-Gaussian statistics!

$[e^{-\int \gamma(t')dt' + i\omega(t-t_0)}]$ is non-Gaussian even when $\gamma(t)$ is Gaussian.]

- $\gamma(t)$ real scalar; two-state Markov jump process

$$\begin{aligned} u(t) = & e^{(-d^+ + i\omega)(t-t_0)} u_0 + \int_{t_0}^t f(s) e^{(-d^+ + i\omega)(t-s)} ds + \sigma \int_{t_0}^t e^{(-d^+ + i\omega)(t-s)} dW(s), \\ u(t) = & e^{(-d^- + i\omega)(t-t_0)} u_0 + \int_{t_0}^t f(s) e^{(-d^- + i\omega)(t-s)} ds + \sigma \int_{t_0}^t e^{(-d^- + i\omega)(t-s)} dW(s). \end{aligned}$$

5. Conditionally linear **SPDE** with regime switching

Spatial extended system:

$$\frac{\partial u(x, t)}{\partial t} = \mathcal{P} \left(\frac{\partial}{\partial x} \right) u(x, t) - \Gamma \left(\frac{\partial}{\partial x} \right) u(x, t) + f(x, t) + \sigma(x) \dot{W}(t).$$

Assume spatial-periodic domain such that

$$u(x, t) = \sum_{k=-\infty}^{\infty} \hat{u}_k(t) e^{ikx}, \quad \hat{u}_{-k}(t) = \hat{u}_k^*(t),$$

$$\sigma(x) \dot{W}(t) = \sum_{k=-\infty}^{\infty} \sigma_k \dot{W}_k(t) e^{ikx}, \quad \sigma_{-k}(t) = \sigma_k^*, \quad W_{-k} = W_k^*,$$

$$\mathcal{P} \left(\frac{\partial}{\partial x} \right) e^{ikx} = \tilde{p}(ik) e^{ikx}, \quad \Gamma \left(\frac{\partial}{\partial x} \right) e^{ikx} = \gamma(ik) e^{ikx},$$

$$\tilde{p}(ik) = i\omega_k, \quad \text{with } -\omega_k \text{ the real valued dispersion relation,}$$

$$\gamma(ik) > 0 \quad \text{represents both explicit and turbulent dissipative processes.}$$

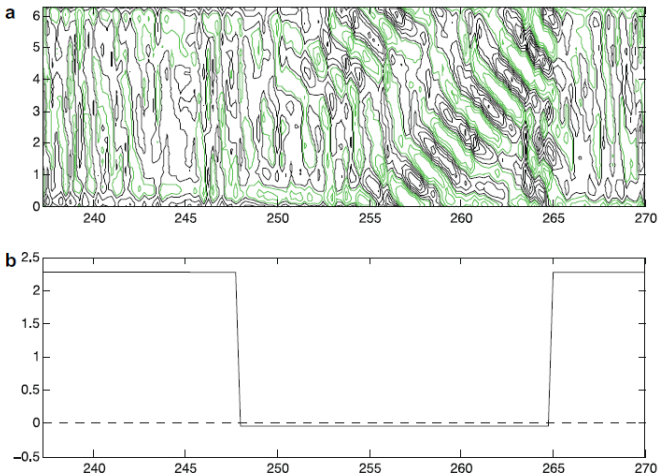
For each Fourier mode, we have

$$d\hat{u}_k(t) = (\tilde{p}(ik) - \gamma(ik)) \hat{u}_k(t) dt + \hat{f}_k(t) dt + \sigma_k dW_k(t), \quad \hat{u}_{-k}(t) = \hat{u}_k^*(t),$$

Use stochastic damping, phase and forcing to model the weak nonlinearity.

Example: Rossby wave with instability transitions. Total number of modes is $N = 52$ with modes 3-5 being intermittent.

$$d\hat{u}_k(t) = (\tilde{p}(ik) - \gamma(ik))\hat{u}_k(t)dt + \hat{f}_k(t)dt + \sigma_k dW_k(t).$$



More examples will be shown in **Lecture 4 and 10-12.**

6. Summary

- Two-state Markov jump process with switching rate $\nu : S_{St} \rightarrow S_{Un}$, $\mu : S_{Un} \rightarrow S_{St}$. The expectation is given by

$$E[X] = \frac{\mu S_{St} + \nu S_{Un}}{\nu + \mu},$$

and the **master equation** has the form

$$\frac{\partial}{\partial t} \hat{P}_t = \hat{P}_t A \quad \text{where} \quad A = \begin{pmatrix} -\nu & \nu \\ \mu & -\mu \end{pmatrix}.$$

- Conditionally linear SDE with regime switching

$$du = [-\gamma(t)u + i\omega u + f(t)]dt + \sigma dW(t).$$

Non-Gaussian statistics if $\gamma(t)$ is a function of time!

- Conditionally linear **SPDE** with regime switching

$$\frac{\partial u(x, t)}{\partial t} = \mathcal{P} \left(\frac{\partial}{\partial x} \right) u(x, t) - \Gamma \left(\frac{\partial}{\partial x} \right) u(x, t) + f(x, t) + \sigma(x) \dot{W}(t).$$

For each Fourier mode, we have

$$d\hat{u}_k(t) = (\tilde{\rho}(ik) - \gamma(ik))\hat{u}_k(t)dt + \hat{f}_k(t)dt + \sigma_k dW_k(t), \quad \hat{u}_{-k}(t) = \hat{u}_k^*(t),$$

Use stochastic damping, phase and forcing to model the weak nonlinearity.

Example: Rossby wave with instability transitions ...

Thank you! Questions? Comments?

References and acknowledgements

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5. We gratefully acknowledge that part of these lecture notes is adapted from Nan Chen's original materials and other online resources.